

MUHAMMAD TAHIR

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PROFESSIONAL SUMMARY

Backend-focused Software Engineering student at MUET with **4 production-grade Java projects** demonstrating deep expertise in **Spring Boot, REST API design, Spring Security, Spring Data JPA**, and **MySQL/JDBC**. Proven ability to architect layered systems with strict separation of concerns, enforce SQL injection prevention via PreparedStatements, and deliver full CRUD lifecycles with proper HTTP semantics. Actively seeking a **Software Engineering or QA Internship** to contribute backend development and API testing skills to a product team.

TECHNICAL SKILLS

Languages: Java (Advanced) · C++ (Intermediate) · SQL (Advanced) · HTML5 · CSS3 · JavaScript
Frameworks: Spring Boot · Spring MVC · Spring Data JPA · Spring Security · Spring Actuator · Servlets · JSP
Databases: MySQL · JDBC (PreparedStatements, Transaction Mgmt) · H2 Embedded Database
Tools: Git/GitHub · Maven · Postman · Apache Tomcat · IntelliJ IDEA · Eclipse IDE · VS Code
Concepts: OOP · MVC Architecture · DAO/DTO Patterns · REST API Design · JUnit · Agile · Software Testing · API Testing

EDUCATION

B.E. in Software Engineering

Mehran University of Engineering & Technology (MUET), Jamshoro, Sindh
Aug 2023 – Aug 2027 (Expected)

Coursework: OOP, Data Structures, Database Systems, Software Engineering, Web Technologies, Discrete Mathematics

Intermediate in Pre-Engineering

AIMS School and College, Matli, Sindh | 2019 – 2021

PROJECTS

Social Media Backend API | Java · Spring Boot · Spring Data JPA · ModelMapper · Lombok · Spring Validation

Jan 2026 – Feb 2026 github.com/Tahir-Dars/SocialMedia

- Architected scalable Spring Boot backend with DTO/Entity separation via ModelMapper, decoupling API contracts from internal domain models — a key production best practice for API stability and versioning
- Enforced controller-layer input validation using Spring Validation annotations, intercepting 100% of malformed payloads before reaching service or persistence layers
- Integrated Lombok annotations eliminating boilerplate getters, setters, and constructors, reducing model class LOC by **40%** and improving long-term maintainability
- Implemented Spring Actuator observability hooks for real-time health metrics and diagnostic endpoints; applied SOLID principles throughout (SRP, DI, IoC)

Banking Customer REST API | Java · Spring Boot · Spring Data JPA · Spring Security · H2 · Maven

Jan 2026 github.com/Tahir-Dars/BankCustomerAPI

- Architected RESTful API for full banking customer lifecycle (CRUD) using a 4-tier Spring Boot structure: Controller → Service → Repository → Model with clean layer boundaries
- Leveraged Spring Data JPA repositories to eliminate boilerplate SQL by **60%** vs equivalent raw JDBC, cutting development time and reducing error surface area
- Integrated Spring Security authentication and Spring Actuator health monitoring, establishing endpoint protection and production observability from project inception
- Delivered complete API with proper HTTP semantics (200, 201, 400, 404, 500) and consistent JSON response contracts across all endpoints

Full-Stack To-Do List Web Application | Java · Servlets · JSP · JDBC · MySQL · Maven · Tomcat · Bootstrap 5

Nov 2025 – Dec 2025 github.com/Tahir-Dars/todolistUsingServlet

- Shipped end-to-end task management app: designed MySQL schema, built 3 HTTP Servlet endpoints (CRUD), deployed WAR to Apache Tomcat, and integrated Bootstrap 5 responsive UI
- Designed 4-layer architecture (Controller → DAO → DB Connector → Entity Model) achieving clean separation of HTTP handling from data access logic with zero inter-layer coupling
- Applied JSTL for server-side rendering and JDBC PreparedStatements across all queries, reducing SQL injection attack surface to zero; verified full CRUD lifecycle across all major browsers

Banking System Application – JDBC Transaction Management | Java · JDBC · MySQL · PreparedStatement

Mar 2025 – Apr 2025 github.com/Tahir-Dars/BankingSystem_JDBC

- Engineered production-grade console banking app with account creation, deposit/withdrawal, and fund transfer — implementing explicit JDBC transaction control with manual commit/rollback on all balance mutations
- Achieved **ZERO data corruption** across **50+ test scenarios** by enforcing atomic transaction boundaries; validated that partial-transfer failures roll back correctly under all tested conditions
- Designed 4-class architecture (BankingApp, Accounts, AccountManager, User) with strict separation of concerns, enabling independent module testing and zero coupling between business logic and persistence layer
- Built secure authentication using PreparedStatements across all database operations, eliminating SQL injection vectors entirely in all 3 transaction types

ACADEMIC PROJECT

Vehicle Rental System | C++, OOP, File I/O | 2024 | MUET Semester Project

- Built console-based vehicle rental system applying OOP pillars — Encapsulation, Inheritance (rental transaction hierarchy), and Polymorphism (vehicle type abstractions) — across inventory, registration, and transaction modules
- Implemented file-based data persistence using C++ file I/O streams for customer and vehicle records, demonstrating data persistence beyond in-memory structures
- Added comprehensive input validation and error handling, reducing invalid-entry failures to zero across 50+ test cases

KEY ACHIEVEMENTS

- **Rapid Technical Progression:** Advanced from raw JDBC/console apps to Spring Boot REST APIs with Security and Actuator integration within **12 months** of self-directed learning
- **Production-Quality Portfolio:** Delivered 4 complete, publicly documented GitHub projects meeting professional architecture standards — layered design, security, and test coverage in every project
- **Data Integrity:** Achieved ZERO data corruption across 50+ banking system test scenarios via explicit JDBC transaction boundaries and atomic operation validation
- **Security-First Development:** Applied SQL injection prevention (PreparedStatements) across 100% of database operations in all projects, with demonstrated understanding of OWASP injection vulnerabilities

LANGUAGES

English — Professional Proficiency

Urdu — Native

Sindhi — Native